MH1301 Discrete Mathematics

Handout 0 - Course Overview

Welcome to MH1301!

Course Information

- Textbook: Kenneth H. Rosen: **Discrete Mathematics and Its Applications**, (any edition, preferably 7th or later), McGraw Hill.
- Reference: Susanna S. Epp: **Discrete Mathematics with Applications**, Thomson.

Textbook is

- Useful for more examples and exercises
- Includes answers to odd-numbered exercises.
- Different levels of difficulty

Course Infomation

- Topics for Part 1: Counting (Weeks 1–4)
  - Basics of Counting (Section 6.1 in textbook)
  - Pigeonhole Principle (Section 6.2)
  - Permutations and Combinations (Section 6.3)
  - Inclusion-Exclusion Principle (Sections 8.5, 8.6)
  - Binomial Coefficients (Sections 6.4–6.6)

- Topics for Part 2: Recurrence Relations (Weeks 4–6)
  - Linear Homogeneous Recurrence Relations (Sections 8.1-8.2)
  - Linear Non-Homogeneous Recurrence Relations (Sections 8.1-8.2)

- Topics for Part 3 (Weeks 7–9, 11–13)
  - Graphs (Sections 10.1–10.8)
  - Trees (Sections 11.1–11.5)

Course Information: Schedule

- Lecturer: Chan Song Heng
  - Email: chansh@ntu.edu.sg
  - Office: SPMS-MAS-04-13
  - Phone: 6513-7453
- Lectures, Weeks 1–13.
  - Time: Monday 3:30pm - 5:20pm, Weeks 1-13, Venue: LT23
  - Online video to makeup for Public holiday on 12 February (Week 5).
- Tutorials, Week 2-13
  - T1 TH 11:30-12:20 SPMS-TR+12
  - T2 TH 11:30-12:20 SPMS-TR+13
  - T3 TH 11:30-12:20 SPMS-TR+17
  - T4 TH 12:30-13:20 SPMS-TR+14
  - T5 TH 12:30-13:20 SPMS-TR+15
  - T6 TH 13:30-14:20 SPMS-TR+12
  - T7 TH 13:30-14:20 SPMS-TR+13
  - T8 TH 13:30-14:20 SPMS-TR+17
  - ~~T9 TH 16.30-17.30 SPMS-TR+13~~

Tutorials

- About the Tutorials
  - Physical classes, 1 hour per week.
  - Tutorial sheet will be uploaded to NTULearn, should have enough time for you to work through before you attend tutorial classes.
  - Tutorial solution will be uploaded after tutorial sessions.
  - To discuss tutorial problems, clarify doubts, check your work, alternative solutions, etc.

- Tutorial participation (weighs 4%)
  - Presenting your solution.
  - Tutor to award for participation (regardless of quality of solution)
  - First three presentations will earn you ~~2% + 1.5% + 0.5%.~~

    2.5% + 1.5%

Course Evaluation

- Weekly Lecture Quizzes (16%)
  Weekly 5-10 minutes quizzes to be given during lectures on weeks 2, 3, 4, 6, 7, 8, 9, 11, 12, and 13.
  Each quiz is worth 2%, the best 8 quiz scores constitutes the 16% component. The lowest 2 scores are dropped.
  You may discuss, refer to notes, etc, for these quizzes.
- Mid-Semester Quiz (20%) on Monday 25 March confirmed
  3:40pm–5:10pm.
  Venue: LT-2A (confirmed).
  Topics: Lectures up to 18 March and Tutorials 1–8.
  Allowed 1 side of an A4-size paper as help sheet.
- Tutorial participation (4%)
- Final Exam (60%) on 9 May 9:00am–11:00am
  Venue: Please check at a later date.

More about the Weekly Lecture Quizzes

- There are no makeups.
- If you miss any weekly quiz with valid reasons, we use the following formula to calculate your 16% weekly quiz component.

No. of quizzes missed **with valid reasons**	Formula
0 or 1	Best 8 quiz scores, worth 2% each
2	Best 7 quiz scores, worth 2.286% each
3	Best 6 quiz scores, worth 2.667% each
4 or more	Best 5 quiz scores, worth 3.2% each

If you have any disagreements, drop me an email.

until all his hardware fail XD

- The plan is to conduct lectures in hybrid mode, physically in LT23 and also on zoom. The quizzes will be on wooclap. So you may take the quizzes without being physically present in the LT.

IMPORTANT: Absence for Mid-Semester Quiz

- Students absent from the Mid-Semester Quiz without valid Leave of Absence will be given zero mark for the missed assessment

  Apply through sPMS

- If you have a valid Leave of Absence for a test, inform the Lecturer within 24 hours and submit official form for leave of absence to your school. own sch

- There will be ONE makeup Mid-Semester Quiz for those who are absent with valid reasons. It will be on the Monday after the actual test Makeup test date: 1 April 2024, 18:00–19:30.

  You will need another valid leave if you are absent for the Makeup Mid-Semester Quiz. The 20% weighting will be shifted to other parts of the assessment if you miss both the actual and the Makeup Quiz with valid reasons.

Academic Integrity

NTU Academic Integrity Policy

- https://www.ntu.edu.sg/wkwsci/admissions/useful-links/undergraduate/academic-integrity
- Plagiarism: (to use as one's own, ideas of another, without acknowledging)
- Academic Fraud: (cheating, collusion, lying and stealing)
  - Bringing in unauthorized material to test and exams
  - Continue writing at the end of test/exam.
  - etc
- Facilitating Academic Dishonesty: (allowing another student to copy an assignment that is supposed to be done individually, allowing another student to copy answers during an examination/assessment, taking an examination or doing an assessment for another student.)

Advice

- Set your objectives
  - To learn and master relevant topics
  - To strengthen foundation of Mathematics
- Manage time
  - Set priority, be disciplined
- Clear your doubts as soon as possible
  - Do not accumulate doubts
  - Discuss with classmates
  - Contact the Lecturer or your Tutor
- Be responsible
  - Read lecture slides or textbook
  - Attempt tutorial problems, attend classes
  - Don't get behind schedule.
- Be helpful in a POSITIVE way
  - Help, encourage and support your classmates to learn
  - Explain to them, discuss with them. It strengthens your own understanding.

Why Discrete Math?

Why study math topics that are “not-continuous”? Why study them separately from continuous mathematics topics such as calculus and analysis?

- Evaluating sums is quite different from evaluating integrals.
- Almost all objects in our daily life are discrete.
- Topics are closely related to Basics of Computer Science
  - data structure
  - algorithms
  - graphics
  - database
  - etc.

Representative Questions

distinct books

Question

How many ways are there to distribute 5 books to 3 students? $3^5 = 243$

Question

How many solutions does the equation $x_1 + x_2 + x_3 = 10$ have, where x_1, x_2, x_3 are nonnegative integers?

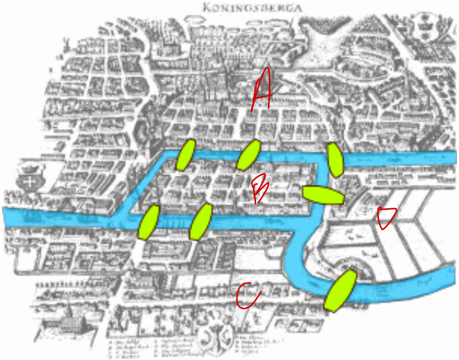
Question

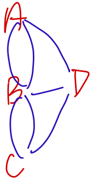
What's the minimum number of steps needed to move n disks from one bar to another in the Tower of Hanoi? $2^n - 1$

Representative Questions

Königsberg Bridges

The Königsberg bridge problem asks if each of the 7 bridges of the figure can be visited exactly once and back to the origin.





Representative Questions

Shortest Path

How to compute the shortest path from one place to another place on the map?